

CATHERINE OLLERMANN

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Petroleum Engineer currently pursuing a Master's degree in Artificial Intelligence. I possess an engineering background in the energy sector and practical experience with Python, SQL, AWS, Power BI, Excel, and data analysis to solve complex problems, support decision-making, and generate business insights. I offer strong analytical and problem-solving capabilities, along with multidisciplinary skills and experience in technical projects and research. Bilingual (English and Portuguese) ready to contribute in technology, data, engineering, and energy roles.

EDUCATION

M.S. in Artificial Intelligence | Atlantis University

Present

Graduate Studies in Science & Technology Management (Computer Science Focus) | Avila University

2025-2026

B.S. in Petroleum Engineering | Federal University of Pelotas

2022

Skills: AWS (EC2, S3, IAM) | Python | SQL | Machine Learning | Pandas | Scikit-learn | Streamlit | Plotly | Neural Networks | Reinforcement Learning | Cloud Infrastructure | Microsoft 365 | Systems Design | Data Analysis | Excel (Data Tracking & Reporting) | Dashboards | Business Analysis | KAPPA Workstation | PMBOK | ALFAsim | CMG Software | tNavigator

TECHNICAL PROJECTS – DATA, AI & CLOUD

[Energy Intelligence Platform](#) | **Data Analytics & Machine Learning** |

2026

- Built and deployed an interactive Python/Streamlit dashboard analyzing 2.5M+ U.S. energy records, with dynamic state comparisons, geographic visualization, statistical insights, and data filtering.
- Developed and evaluated ML trend models using Scikit-learn and temporal validation; deployed an interactive public demo with Plotly visualizations and optional CSV analysis.

Avila University | Sentiment Analysis with Machine Learning

2026

- Developed an end-to-end sentiment analysis pipeline using Python, Hugging Face Transformers, PyTorch, and Scikit-learn to classify customer reviews.
- Performed data preprocessing, model training, evaluation, and visualization of performance metrics, applying Natural Language Processing (NLP) techniques to generate actionable insights.

Avila University | Cloud Infrastructure on AWS

2025

- Designed and deployed a cloud infrastructure environment on AWS, configuring core services such as EC2, S3, and IAM to support scalable and secure application environments.
- Applied cloud architecture best practices, including identity and access management, resource organization, and basic cost optimization strategies to ensure efficiency and system reliability.

PROFESSIONAL EXPERIENCE

Gas Energy Chemicals | Business Analyst Intern

01/2021 – 06/2021

- Maintained monthly datasets tracking production volumes, oil and gas pricing, and consumption metrics across sectors, ensuring data accuracy and consistency for internal use.
- Developed recurring Excel reports and dashboards consolidating key operational and market data, enabling visibility into performance trends and supporting ongoing business and project analysis.
- Analyzed operational and market data to identify trends and variances, contributing to the development of internal studies and client deliverables, including projects within the Energy and Oil & Gas sectors.

Federal University of Pelotas | Engineering Tutor

12/2020 – 06/2021

- Guided students through problem-solving exercises and technical concepts, providing instruction in ALFAsim software, improving students' ability to apply simulation tools in academic projects.
- Delivered academic support in Applied Economics, breaking down complex concepts into practical examples and supporting students in assignments and coursework to strengthen comprehension and performance.